# Representation, Not Metaheuristic, Governs Feasibility-Preserving Multi-Objective Optimization: A Study on United States Employment-Based Visa Allocation

Javier Augusto Rebull Saucedo, Yazmín Ivonne Flores Martínez, and Raúl Gibrán Porras Alaniz

Universidad Autónoma de Ciudad Juárez,
Maestría en Inteligencia Artificial y Analítica de Datos,
Ciudad Juárez, Chihuahua, México
rebull@outlook.com

**Abstract.** Many hard constrained combinatorial problems are solved by pairing a metaheuristic with a feasibility-preserving decoder, yet the true driver of performance is seldom isolated. Across *four* structurally distinct multi-objective problems, performance is governed far more by the match between representation and search operators than by the metaheuristic family. We develop this on a real case: allocating roughly 140,000 U.S. employment-based visas per year under a per-country ceiling and a spillover cascade, cast as a three-objective integer program (waiting load, inter-country disparity, and visa waste) under six hard constraints and solved with multi-objective Harris Hawks Optimization (MOHHO). Its core is a smallest-position-value random-key mapping composed with a greedy assignment that satisfies every constraint by construction—no penalties, no repair. A controlled six-method ladder isolates the cause: the decoder alone (random restart) already beats a real-coded NSGA-II, and matching operators to the representation lifts three distinct paradigms (genetic algorithm, decomposition, and swarm) above every random-key method—our representation-matched Discrete-MOHHO being the most stable. We stress-test this against a search-space-collapse confound—non-saturating decoders, reference-point sweeps, and an exact bi-objective MILP front—and find the representation effect robust though amplified by the decoder's budget saturation. Replicated on a knapsack, a traveling salesman, and a flow-shop, a matched operator is the best optimizer on *every* structure, while the collapse of a tuned GA below random sampling proves specific to selection landscapes. Parameters are fixed by a Taguchi design; the recovered front Pareto-dominates the incumbent first-in, first-out policy chiefly by eliminating visa waste, yielding an auditable menu of trade-offs as decision support, not a single recommendation.

**Keywords:** Multi-objective optimization · Harris Hawks Optimization · Constraint handling · Random-key decoder · Permutation encoding · Taguchi design of experiments · Resource allocation

## 1 Introduction

Each fiscal year the United States Congress authorizes approximately 140,000 employment-based (EB) immigrant visas, distributed across five categories (EB-1 to EB-5). Two structural rules govern the distribution: a *per-country ceiling* ($P_c \approx 25{,}620$ visas, the statutory 7 % cross-preference limit) that no single country may exceed, and a *spillover* mechanism that cascades unused visas across categories (EB-4/5 $\rightarrow$ EB-1 $\rightarrow$ EB-2 $\rightarrow$ EB-3). The incumbent system processes petitions strictly first-in, first-out (FIFO) by priority date. The interaction of these rules makes the incumbent policy dysfunctional: two countries concentrate roughly three-quarters of demand (India $\approx$ 63 %, China $\approx$ 14 %) and accumulate waits of 10–13 years, while the clash between per-country and per-category ceilings leaves thousands of visas unused.

This matters beyond accounting. Behind every backlogged petition is a household whose relocation, employment, and family reunification hinge on a priority date, so the allocation rule has direct welfare consequences. The problem admits three legitimate, mutually conflicting goals: serve the longest-waiting petitions (*efficiency*), distribute opportunity evenly across nationalities (*equity*), and waste no visas (*utilization*). No single allocation optimizes all three; the policy-relevant object is the set of best compromises—a Pareto front.

Computing that front is hard for three reasons. First, the allocation is *combinatorial*: it assigns integer visa counts to 105 country$\times$category groups under six simultaneous constraints, and its feasible set generalizes the multidimensional knapsack problem, which is NP-hard [12]. Second, the objectives are genuinely *conflicting*, so the solution is a front rather than a point. Third—and most often underestimated—the *hard constraints* are unforgiving: a continuous population metaheuristic naturally proposes infeasible candidates, and the usual remedies (penalty functions, aggressive repair) either require delicate tuning or distort the very search landscape the metaheuristic explores.

We address these challenges with a multi-objective adaptation of Harris Hawks Optimization (MOHHO) whose centerpiece is a *feasibility-preserving decoder*. Our central contribution is general: across four structurally distinct problems, the *representation–operator match*—not the metaheuristic family—governs decoder-based search, with the U.S. visa problem as the lead case study. Concretely, this paper makes five contributions: *(i)* a decoder that couples a smallest-position-value (SPV) random-key mapping with a deterministic greedy assignment, guaranteeing all six policy constraints *by construction*—without penalties or repair, and therefore without perturbing the search landscape; *(ii)* a multi-objective HHO that pairs this decoder with a crowding-pruned external archive and applies it to a real, constraint-heavy allocation problem rather than to synthetic benchmarks; *(iii)* a Taguchi L9 orthogonal-array tuning of the population size and iteration budget, replacing empirical parameter choices with a systematic experimental design validated by a confirmation run; *(iv)* a controlled six-method ladder that, by lifting three distinct paradigms (a genetic algorithm (GA), decomposition, and a representation-matched *Discrete-MOHHO* we introduce) to within about 1 % of one another once operators match the representation, shows that the feasibility-preserving decoder and the representation–operator match, not the choice of metaheuristic, govern performance—with a direct operator-order-preservation measurement that explains *why* (interpolating GA operators are near-identity on the decoded permutation); and *(v)* a rigorous empirical protocol (30 runs, omnibus Friedman/Nemenyi tests with effect sizes, domination of FIFO, robustness across five perturbed-demand instances, and a four-structure generalization—knapsack, traveling salesman, and flow-shop—that separates the transferable representation–operator law from the selection-specific GA-below-random collapse) that makes every claim reproducible.

## 2 Background and Related Work

Harris Hawks Optimization (HHO) [13] is a population-based metaheuristic whose prey *escape energy* $E$ drives a continuous exploration–exploitation transition, with optional Lévy-flight dives. It has been extended to multiple objectives via external archives, elitist non-dominated sorting, and grid indices [15,4,14,6]; these variants are validated mostly on continuous benchmarks, leaving their use on *constrained combinatorial* allocation—where feasibility, not convergence, is the binding difficulty—comparatively unexplored.

We bridge continuous search and combinatorial structure with random keys [2]: a real vector is decoded to a feasible permutation by sorting (the smallest-position-value, SPV, rule). This idea underlies biased random-key genetic algorithms [16] and a general random-key optimizer that pairs a decoder with *any* metaheuristic [5]—the paradigm we instantiate for HHO and probe by varying the metaheuristic while holding the decoder fixed. Among constraint-handling families [18], penalty methods need problem-specific weights and blur feasibility, repair can collapse diversity, and decoder (feasibility-preserving) approaches guarantee feasibility by construction; we adopt the third. NSGA-II (Non-dominated Sorting Genetic Algorithm II) [9] is our benchmark and the standard tool for hard allocation and scheduling [23]; we tune parameters with a Taguchi orthogonal array [17,1] and assess fronts with the hypervolume [24], inverted generational distance and spacing [7], nonparametric tests [10], and the $A_{12}$ effect size [21].

## 3 Problem Formulation

*Decision variables.* The problem is discretized into $G = C \times |\mathcal{J}| = 21 \times 5 = 105$ *groups*, where each group $g$ is a (country $c$, category $j$) pair. The decision vector is $\mathbf{x} = (x_1, \ldots, x_{105}),\ x_g \in \mathbb{Z}_0^+$, where $x_g$ is the number of visas assigned to group $g$. Each group has a demand (backlog) $n_g$ and a *waiting weight* $w_g = t_{\text{now}} - d_g$, the years elapsed since its priority date $d_g$.

*Objectives.* We minimize three conflicting objectives:

$$f_1(\mathbf{x}) = \frac{\sum_g (n_g - x_g)\, w_g}{\sum_g n_g} \qquad \text{unserved waiting load (efficiency),} \tag{1}$$

$$f_2(\mathbf{x}) = \max_{c_1, c_2 \in \mathcal{C}} \left| \bar{W}_{c_1} - \bar{W}_{c_2} \right| \qquad \text{maximum inter-country disparity (equity),} \tag{2}$$

$$f_3(\mathbf{x}) = V - \textstyle\sum_g x_g \qquad \text{wasted (unassigned) visas (utilization),} \tag{3}$$

where $V$ is the total available visas and $\bar{W}_c = \big(\sum_{g:\, c(g)=c} x_g w_g\big) / \big(\sum_{g:\, c(g)=c} x_g\big)$ is the visa-weighted mean wait of country $c$. Thus $f_2$ measures the gap between the longest- and shortest-waiting nationalities: the smaller $f_2$, the more equitable the distribution. (A country receiving no visas takes its worst backlog wait as $\bar{W}_c$, so $f_2$ implicitly penalizes starving a nationality; FIFO and all reported equity solutions serve every country.)

*Constraints.* The feasible set $\mathcal{F}$ is defined by six hard constraints:

| | |
|---|---|
| R1: $\sum_g x_g \le V = 140{,}000$ | (total visas) |
| R2: $\sum_{g:\, c(g)=c} x_g \le P_c = 25{,}620 \ \ \forall c$ | (7 % per-country ceiling) |
| R3: $\sum_{g:\, j(g)=j} x_g \le K_j^{\text{eff}} \ \ \forall j$ | (effective per-category cap) |
| R4: $0 \le x_g \le n_g \ \ \forall g$ | (do not exceed demand) |
| R5: $x_g \in \mathbb{Z}_0^+ \ \ \forall g$ | (integrality) |
| R6: FIFO order within each $(c, j)$ queue | (seniority within a group) |

where $K_j^{\text{eff}}$ is the per-category cap after the spillover cascade (retained for generality but inert here: every category is oversubscribed, so the slack is zero and $K_j^{\text{eff}} = K_j^{\text{base}}$). The result is a three-objective integer program with six constraints whose feasible space grows combinatorially with $G$, motivating a metaheuristic rather than exact enumeration.

*Pareto optimality.* A solution $\mathbf{x}$ *dominates* $\mathbf{x}'$ ($\mathbf{x} \prec \mathbf{x}'$) iff $f_i(\mathbf{x}) \le f_i(\mathbf{x}')$ for all $i$ and $f_i(\mathbf{x}) < f_i(\mathbf{x}')$ for some $i$. The *Pareto front* is $\mathcal{P}^* = \{\mathbf{x} \in \mathcal{F} \mid \nexists \mathbf{x}' \in \mathcal{F} : \mathbf{x}' \prec \mathbf{x}\}$. MOHHO approximates $\mathcal{P}^*$ through an external archive.

# 4 Methods

## 4.1 Multi-objective Harris Hawks Optimization

Each hawk is a position $\mathbf{H} \in \mathbb{R}^{105}$ in the box $[0,1]^{105}$. At iteration $t$ the escape energy is

$$E = 2E_0\big(1 - \tfrac{t}{T}\big)\,, \qquad E_0 \sim U(-1, 1), \tag{4}$$

which drives the choice among six movement operators selected by $|E|$ and uniform draws $q, r \in U(0,1)$ (Table 1). The multi-objective adaptation introduces

**Table 1.** The six MOHHO movement operators, selected by the escape energy $|E|$ and uniform draws $q, r$. The archive's leader supplies the prey position.

| Op. | Condition | Behavior |
|---|---|---|
| OP1 | $\lvert E\rvert \geq 1,\ q \geq 0.5$ | Exploration from a random hawk in the swarm. |
| OP2 | $\lvert E\rvert \geq 1,\ q < 0.5$ | Exploration relative to the population centroid. |
| OP3 | $0.5 \leq \lvert E\rvert < 1,\ r \geq 0.5$ | Soft besiege: gradual approach to the prey. |
| OP4 | $\lvert E\rvert < 0.5,\ r \geq 0.5$ | Hard besiege: aggressive contraction. |
| OP5 | $0.5 \leq \lvert E\rvert < 1,\ r < 0.5$ | Soft besiege with rapid Lévy dives. |
| OP6 | $\lvert E\rvert < 0.5,\ r < 0.5$ | Hard besiege with rapid Lévy dives. |

three elements. First, the *leader* ("rabbit") is not a single global best but a solution drawn from the external archive with probability proportional to its crowding distance, which biases search toward sparsely populated regions of the front. Second, an *external archive* stores non-dominated solutions and, when it overflows, is pruned by removing the solution with the smallest crowding distance. Third, the archive is updated after every move using the non-dominance relation. The archive size is treated as a tunable parameter (Section 5).

For a function-evaluation-fair comparison, the Lévy operators OP5/OP6 accept a single trial point per hawk per iteration (the canonical HHO evaluates two), so MOHHO consumes exactly $N \times T$ evaluations—identical to every other method.

### 4.2 A feasibility-preserving decoder

The HHO operators act in the continuous space $\mathbb{R}^{105}$, but the problem is combinatorial with six hard constraints. We bridge the two with the SPV rule [2] followed by a deterministic greedy assignment (Algorithm 1). The SPV rule turns a hawk $\mathbf{H}$ into a permutation $\boldsymbol{\pi}$ by sorting the group indices in ascending order of their key $h_g$. The greedy decoder then walks the groups in that order and assigns

$$x_g = \min\bigl(n_g,\ V_{\text{rem}},\ P^{\text{rem}}_{c(g)},\ K^{\text{rem}}_{j(g)}\bigr), \tag{5}$$

decrementing the residual budgets of R1–R3 after each step. Because every assignment is capped by the remaining budgets and by demand, and counts are integers, *every* decoded vector satisfies R1–R6 by construction. No penalty term is introduced and no repair is performed, so the hawks search directly on a feasible objective landscape, preserving HHO's cooperative foraging and stochasticity. The greedy fill is budget-saturating (each visited group gets its maximal feasible amount), so the decoder's image is the saturating subset of $\mathcal{F}$ and all dominance claims are relative to it—the right object here, since minimizing waste $f_3$ already makes reserving budget unattractive.

**Proposition 1 (Feasibility by construction).** *For any hawk $\mathbf{H} \in \mathbb{R}^{105}$, the allocation $\mathbf{x}$ returned by Algorithm 1 satisfies all six constraints R1–R6.*

**Algorithm 1** Feasibility-preserving SPV + greedy decoder

**Require:** hawk $\mathbf{H} \in \mathbb{R}^{105}$; demands $n_g$; budgets $V$, $\{P_c\}$, $\{K_j^{\text{eff}}\}$

$\boldsymbol{\pi} \leftarrow \textsc{ArgSortAscending}(\mathbf{H})$ ▷ SPV: keys → permutation

$V_{\text{rem}} \leftarrow V$; $P_c^{\text{rem}} \leftarrow P_c\ \forall c$; $K_j^{\text{rem}} \leftarrow K_j^{\text{eff}}\ \forall j$

**for** $g \in \boldsymbol{\pi}$ **do** ▷ visit groups in SPV order

  $x_g \leftarrow \min\big(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}}\big)$

  $V_{\text{rem}} \mathrel{-}= x_g$; $P_{c(g)}^{\text{rem}} \mathrel{-}= x_g$; $K_{j(g)}^{\text{rem}} \mathrel{-}= x_g$

**end for**

**return x** ▷ satisfies R1–R6 without repair

*Proof.* The residuals are initialized to $V$, $P_c$, $K_j^{\text{eff}} \geq 0$ and, at each step, decremented by $x_g = \min(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}})$. By induction each residual stays $\geq 0$, since $x_g$ never exceeds the current residual; hence $0 \leq x_g \leq n_g$ (R4) and, as the minimum of nonnegative integers, $x_g \in \mathbb{Z}_0^+$ (R5). Because $x_g \leq V_{\text{rem}}$ and $V_{\text{rem}}$ is the unconsumed total, $\sum_g x_g \leq V$ (R1); likewise $x_g \leq P_{c(g)}^{\text{rem}}$ and $x_g \leq K_{j(g)}^{\text{rem}}$ keep the per-country and per-category sums within $P_c$ (R2) and $K_j^{\text{eff}}$ (R3). Finally, each group is a single $(c, j)$ pair sharing one priority date, so assigning the count $x_g$ serves its $x_g$ most-senior petitions, honoring FIFO order within the queue (R6). Thus $\mathbf{x} \in \mathcal{F}$ for every input, with no penalty term and no repair. □

### 4.3 Benchmarks and a representation-matched optimizer

To separate the contribution of the decoder, the representation, and the search heuristic, we compare a *ladder* of methods that all share the same greedy decoder, objectives, hypervolume reference point, and $50 \times 500$ budget; they differ only in how they explore. Two operate on the continuous SPV random keys in $\mathbb{R}^{105}$: *(a)* the MOHHO of Section 4.1; and *(b)* **NSGA-II** [9] with simulated binary crossover (SBX, $\eta_c = 20$, $p_c = 0.9$), polynomial mutation ($\eta_m = 20$, $p_m = 1/d$), binary tournament selection, and elitist $(\mu+\lambda)$ replacement. A third, *(c)* **random restart**, draws the 25,000 candidates uniformly at random (no search), isolating what the decoder alone achieves.

The ablation (Section 6.3) reveals that the continuous random-key operators are a poor match for the induced permutation landscape. We therefore also evaluate three **permutation-native** methods—spanning three distinct paradigms—that act directly on permutations of the 105 groups (decoded by the same greedy procedure): *(d)* **permutation-NSGA-II** (dominance-based) with order crossover (OX) and swap mutation; *(e)* **permutation-MOEA/D** [22] (decomposition-based), with Tchebycheff scalarization over a simplex-lattice of weight vectors and the same OX + swap operators; and *(f)* **Discrete-MOHHO**, our representation-matched Harris Hawks optimizer. Discrete-MOHHO keeps HHO's signature—the escape energy $E$ of Eq. (4) scheduling exploration and exploitation, and a crowding-pruned external archive supplying the leader—but expresses every move on permutations: when $|E| \geq 1$ a hawk explores by recombining (OX) with a random hawk or by a heavy random shuffle; when $|E| < 1$ it besieges the leader by inheriting an order segment from it (OX) and perturbing with a number of swaps that shrinks as $|E| \to 0$, with occasional Lévy-like segment reversals. All six methods run for 30 independent seeds under the identical budget.

### 4.4 Evaluation protocol and indicators

We assess each approximation front with four indicators: the *hypervolume* (HV; larger is better) dominated with respect to the reference point (10; 16; 20,000), an upper bound on each objective; the *inverted generational distance* (IGD; smaller is better) to a reference front $Z$ formed by the non-dominated union of the MOHHO and NSGA-II combined fronts ($|Z| = 113$); the *spacing* indicator (smaller is better) measuring uniformity; and the *cardinality* of the non-dominated set. Objectives are min–max normalized before IGD and spacing. Each algorithm is run for 30 independent seeds, and a *combined front* is formed as the non-dominated union of its 30 per-run fronts. Statistical significance between the two 30-value HV distributions is assessed with a one-sided Mann–Whitney $U$ test [10], and effect size with the Vargha–Delaney $A_{12}$ [21]. The deterministic **FIFO** baseline orders groups by ascending priority date and is decoded with the same greedy procedure, yielding $f_1 = 8.7891$, $f_2 = 13.0$ years, and $f_3 = 1{,}940$ wasted visas.

## 5 Parameter Tuning by Taguchi Orthogonal Array

We replace empirical parameter choices by a Taguchi L9($3^4$) orthogonal-array design [17,1] over four factors at three levels—population $N \in \{30, 50, 70\}$, iteration budget $T \in \{100, 300, 500\}$, archive size $\in \{50, 100, 150\}$, and Lévy exponent $\beta \in \{1.3, 1.5, 1.7\}$—with the larger-the-better signal-to-noise (S/N) ratio of the hypervolume as response, each of the nine configurations replicated over 12 seeds disjoint from the evaluation seeds.

Averaging the S/N ratio per factor level (Table 2, Figure 1) gives an unambiguous ranking: *population* ($\Delta = 0.318$ dB) and *iteration budget* ($\Delta = 0.155$ dB) dominate, while $\beta$ (0.108) and archive size (0.046) lie within run-to-run noise; the population effect grows toward the top of its range while the iteration effect saturates past $T$=300. The additive model predicts the optimum $N$=70, $T$=500 at S/N $\approx$ 109.78 dB; a confirmation run gives 109.66 dB (HV = 304,391), validating the model. The 30-run study adopts $N$=50, $T$=500, archive 100, $\beta$=1.5, whose confirmation gives S/N 109.56 dB—within 1.2 % hypervolume of the L9 optimum (its sole difference is $N$=50 vs. 70), a deliberately conservative, compute-economical operating point ($T$=500 far exceeds the iteration-138 convergence point).

## 6 Results

### 6.1 Combined Pareto front and domination of FIFO

Combining and re-filtering the 30 runs yields a front of **92 non-dominated solutions**. Figure 2 shows it in the full objective space and Figure 3 in the $f_1$–$f_2$

**Table 2.** Response table: mean S/N (dB) per factor level, range $\Delta$, and rank. Larger $\Delta$ means a more influential factor; the highest-S/N level in each row is in bold. Each $\Delta$ is computed from unrounded S/N values.

| Factor | Level 1 | Level 2 | Level 3 | $\Delta$ | Rank |
|---|---|---|---|---|---|
| A: population $N$ | 109.314 | 109.443 | **109.632** | 0.318 | 1 |
| B: iteration budget $T$ | 109.372 | 109.489 | **109.528** | 0.155 | 2 |
| C: archive size | 109.461 | **109.487** | 109.441 | 0.046 | 4 |
| D: Lévy exponent $\beta$ | 109.462 | **109.518** | 109.410 | 0.108 | 3 |

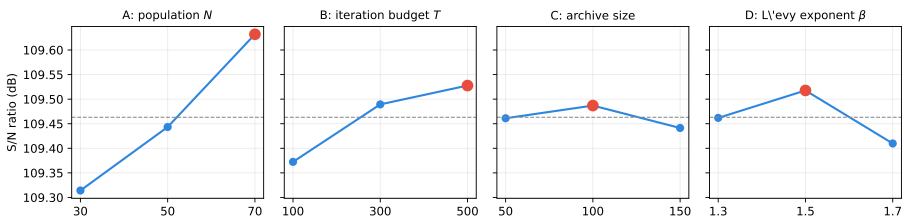


**Fig. 1.** Main-effects plot of the S/N ratio. The dashed line is the grand mean (109.46 dB) and the red marker the best level per factor; population and iteration budget show clear trends (population the steeper), while archive size and $\beta$ are essentially flat.

projection; in both, the FIFO baseline floats *above* the front, visual evidence that it is dominated. The extreme solutions (Table 3) expose the trade-offs.

The central empirical finding is that **FIFO is not Pareto-optimal**: the minimum-$f_1$ solution Pareto-dominates it—strictly better on waste ($f_3$ from 1,940 to 680 visas), tied on the maximal disparity ($f_2 = 13.0$), and only negligibly better on the near-degenerate $f_1$ (8.7891 to 8.7884). The substantive case against FIFO is thus about waste and equity: 82 of the 92 front solutions reach $f_3 = 0$ (zero waste, which FIFO never attains), with disparities spanning from 13.0 years down to 2.0. (The MOHHO, Discrete-MOHHO, and permutation-NSGA-II combined fronts agree within 0.2 % in hypervolume (Table 4), so the policy menu does not hinge on the optimizer choice—the recommended single-run optimizer, Discrete-MOHHO, yields an essentially equivalent front.)

### 6.2 Convergence and robustness

The mean hypervolume reaches 95 % of its final value by iteration 12 and 99 % by iteration 138; gains beyond iteration 140 are under 1 %. Over the 30 runs the HV distribution is narrow (mean 302,379, coefficient of variation [CV] 2.47 %), evidencing a stable, reproducible algorithm.

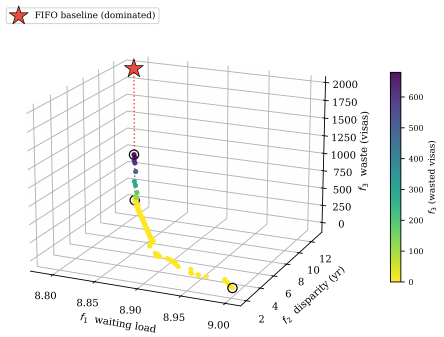


**Fig. 2.** The 92-solution combined Pareto front over the three objectives, colored by the waste $f_3$. The FIFO baseline (red star) floats above the front—the dotted drop-line marks its distance to the front's waste level—and is dominated; circles mark the extreme solutions.

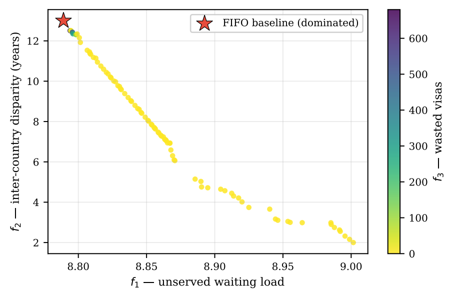


**Fig. 3.** $f_1$–$f_2$ projection of the same front, colored by $f_3$ (wasted visas). FIFO lies inside the dominated region.

**Table 3.** Extreme solutions of the combined front, compared with the FIFO baseline. The minimum-$f_1$ solution dominates FIFO (better on two objectives, equal on the already-maximal disparity).

| **Solution** | $f_1$ | $f_2$ **(yr)** | $f_3$ **(visas)** | **Remark** |
|---|---|---|---|---|
| Min. $f_1$ (least waiting) | 8.7884 | 13.0 | 680 | **Dominates FIFO** |
| Min. $f_2$ (most equitable) | 9.0016 | 2.0 | 0 | Radical equity |
| Min. $f_3$ (full use) | 8.851 | 8.06 | 0 | 100 % utilization |
| FIFO baseline | 8.7891 | 13.0 | 1,940 | Reference (dominated) |

### 6.3 Decoder, representation, and search: a controlled ladder

We attribute performance to its true source by climbing a ladder of methods, all sharing the decoder and a 25,000 function-evaluation budget—the comparison currency (Table 4, Figure 4); the permutation-native methods inherit the Taguchi-tuned $N$ and $T$ and are not separately tuned, so any edge is not a tuning artifact. Under this budget MOHHO beats the real-coded NSGA-II by 3.1 % in mean hypervolume (one-sided Mann–Whitney $p \approx 1.3 \times 10^{-5}$, $A_{12} = 0.82$), with smaller IGD and lower spacing.

*The decoder does most of the work. Random restart*—blind sampling with no search—already reaches a mean HV of 309,821: 5.6 % *above* NSGA-II and, strikingly, 2.5 % *above* classic MOHHO itself (the swarm wins on only 9/30 paired seeds). Under an identical evaluation budget the decoder shapes a landscape on which blind sampling outperforms *both* real-coded searches; giving NSGA-II the *identical* archive barely helps it (293,666 vs. 293,367), so NSGA-II's deficit is not an archiving artifact. Nor is the swarm's loss an artifact of our single-trial-point budgeting: a control with the canonical two-point Lévy dive does not beat random restart either (308,926 at the matched budget, paired Wilcoxon $p = 0.77$; only a *tie* even at its 42 %-over-budget native schedule).

*The representation–operator match is decisive.* Why does a tuned NSGA-II fall below blind sampling? We measured the cause directly. On the SPV random keys, SBX and polynomial mutation are *near-identity on the decoded permutation*: over 3,000 trials the Kendall rank correlation between a parent's SPV order and its child's is $\tau = 0.99$ (and stays in $[0.987, 0.992]$ when measured on-trajectory). Sweeping the crossover spread $\eta_c \in \{2, \ldots, 100\}$ confirms this is no tuning artifact: even the most disruptive setting ($\eta_c$=2, $\tau$=0.92) leaves the GA's mean HV below blind random restart, and across the sweep order-preservation and HV are perfectly anti-correlated (Kendall $-1.0$). The GA perturbs the keys so little that the induced permutation barely changes, leaving it almost frozen in combinatorial space—precisely why it sinks below blind sampling, which at least draws fresh permutations. HHO's position updates, by contrast, induce large permutation jumps ($\tau \approx 0$), so even on random keys the swarm genuinely traverses the decoded space (hence MOHHO > NSGA-II within the random-key tier). The clean fix, though, is to match the operators to the representation. Acting *directly* on permutations lifts *three different paradigms* above every random-key method: permutation-NSGA-II (dominance/GA) reaches 318,151, permutation-MOEA/D (decomposition) 314,846, and our **Discrete-MOHHO** (swarm) 316,637. The three cluster within about 1 % of one another, yet all sit 1.6–2.7 % above the best random-key method (random restart)—more than the spread *among* the matched paradigms. Discrete-MOHHO improves on classic MOHHO by +4.7 % (paired Wilcoxon $p = 9.3 \times 10^{-10}$, better on all 30/30 seeds) and is the *most stable* method (CV 0.54 %, versus 0.58 % for permutation-NSGA-II and 2.47 % for classic MOHHO), making it the most dependable choice for a single-run decision. The marginal mean lead belongs to permutation-NSGA-II (+0.48 % over Discrete-MOHHO, $A_{12} = 0.27$).

*An empirical regularity.* The ladder (Figure 4) makes the pattern unmistakable: performance is governed by the *feasibility-preserving decoder* and the *representation–operator match*, not by the metaheuristic family. That encoding–operator mismatch can hurt search is known in principle since random keys were introduced [2]; our contribution is the *controlled, quantified* demonstration on a real constrained problem—including the counterintuitive result that *both* real-coded searches (a tuned NSGA-II and the MOHHO swarm) fall below blind sampling—and the decomposition of performance into decoder, representation, and family. That three *distinct* paradigms (swarm, decomposition, dominance/GA) converge to within about 1 % once the representation is matched, all far above the random-key methods, is strong evidence that the family is second-order here. We are precise about what this shows: the operator–representation *match* is first-order (the random-key→permutation jump), whereas *within* a matched representation both the operator set and the selection architecture

**Table 4.** The method ladder: 30 runs each, identical greedy decoder and 25,000-evaluation budget. Top block: random-key encoding. Bottom block: permutation-native, spanning three distinct paradigms (swarm, decomposition, dominance/GA), which cluster within about 1 % and dominate every random-key method. Best per column in bold; "perm-" abbreviates "permutation-".

| **Method** | **Paradigm** | **Mean HV** $\pm\sigma$ | **CV** | **Comb. HV** | **# sols** |
|---|---|---|---|---|---|
| NSGA-II | GA | $293{,}367 \pm 6{,}996$ | 2.38 % | 316,060 | 92 |
| MOHHO | swarm | $302{,}379 \pm 7{,}455$ | 2.47 % | 321,446 | 92 |
| Random restart | — | $309{,}821 \pm 2{,}513$ | 0.81 % | 316,383 | 81 |
| Discrete-MOHHO | swarm | $316{,}637 \pm 1{,}720$ | **0.54 %** | 321,354 | **149** |
| perm-MOEA/D | decomp. | $314{,}846 \pm 5{,}384$ | 1.71 % | 320,969 | 90 |
| perm-NSGA-II | GA | $\mathbf{318{,}151 \pm 1{,}855}$ | 0.58 % | **321,935** | 148 |

are second-order—a $3 \times 3$ operator $\times$ architecture factorial (OX+swap, PMX (partially-mapped crossover)+insertion, CX (cycle crossover)+inversion $\times$ the three architectures, 30 seeds) attributes only 5.5 % and 13.4 % of the hypervolume variance to them, against 78 % seed noise. The matched swarm reclaims competitiveness and the best stability, which is why we promote Discrete-MOHHO. An omnibus test on a common 30-seed set—a paired Friedman test with a Nemenyi post-hoc, identical to Section 6.5—rejects equal performance decisively ($\chi^2 = 112.6$, $p \approx 1.2 \times 10^{-22}$, critical difference 1.38); the mean ranks split into two tiers, the three permutation-native methods (1.6–2.5) above the three random-key methods (4.2–5.8), NSGA-II worst, with finer within-tier gaps inside the critical difference. The ranking is also robust to the hypervolume reference point: across five reference points random restart outranks both real-coded searches throughout, and the strict permutation/random-key tier split holds under four of the five, breaking only at the tightest. We frame this as an empirical regularity for this problem, not a general law: all instances here share one structural skeleton—which the structurally distinct problems of Section 6.5 directly address.

*Robustness to a search-space-collapse confound.* A skeptic could argue that the decoder's budget saturation collapses the search space so far that the optimizer is necessarily second-order. We tested this on three axes. *Structure*: 50,000 random permutations decode to 39,401 distinct objective points (degeneration ratio 1.27), and a principal-component analysis (PCA) places about 99 % of the front's variance in two components—the effective problem is bi-objective, not mono-objective, though $f_2$ dominates ($\sim$39$\times$ wider than $f_1$ across these random decodes), a deliberately weak multi-objective lead case. *Decoder*: re-running the full ladder on two *non-saturating* decoders (a fractional-intensity and a stochastic-skip decoder, both feasible by construction) keeps the permutation tier significantly above the random-key tier under *all three* decoders (paired Wilcoxon $p < 10^{-3}$) and never inverts it; the margin shrinks from 6.9 % (greedy) to 4.3 % and 1.4 %, so saturation *amplifies*—it does not create—the effect, and an exact $(f_1, f_3)$ mixed-integer linear program (MILP) front (with $f_2$ a posteriori) exposes

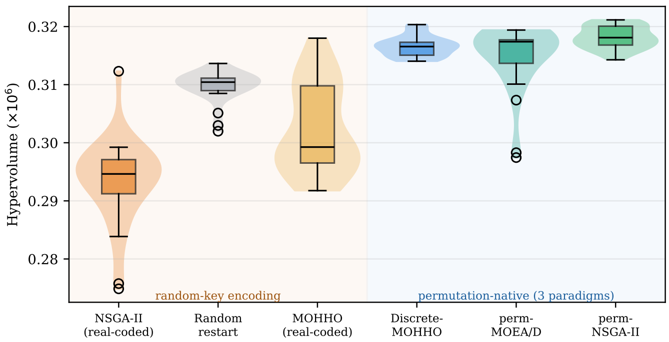


**Fig. 4.** The method ladder: per-run hypervolume (30 runs each, identical decoder and budget). Crossing from random-key encodings (left, shaded amber) to permutation-native operators (right, shaded blue) produces a clear jump; the metaheuristic family (GA vs. Harris Hawks) matters far less than the representation. Discrete-MOHHO is the tightest distribution (most stable).

no practically meaningful non-saturating solution the greedy front misses. *Metric*: the GA operators' near-identity holds on the full decoded *allocation*, not just the order (an SBX step barely perturbs the decoded vector—consistent with its order near-identity $\tau = 0.99$—whereas an HHO step moves it substantially), and the per-run tier ranking is reference-point-robust (above)—though on the *combined* fronts the strict tier softens under reference-free metrics (IGD$^+$, additive $\epsilon$), which converge to a near-common reference set, so the tier is a per-run phenomenon, and the hypervolume that exhibits it is dominated by the $f_2$ axis—the only objective with a wide range, $f_1$ being more than an order of magnitude narrower and $f_3$ saturating. The confound thus *bounds* rather than overturns the claim: the effect is real and robust, modest on this saturation-amplified, $f_2$-dominated instance, with the general law resting on the four structures below.

### 6.4 Generalization across instances

To check robustness to the demand estimate, we built five further instances perturbing every group's demand by an independent uniform $\pm 20\,\%$ (fixed seeds) and recomputing the spillover-adjusted category and per-country caps. On all five—as on the base instance—FIFO is dominated, MOHHO outperforms NSGA-II (one-sided Mann–Whitney $p \leq 0.071$ each), and random restart matches or exceeds MOHHO (101–103 % of its HV) while both stay above NSGA-II (96–99 % of MOHHO). This is robustness to demand perturbation; structural generalization is addressed next.

### 6.5 Generalization across structurally distinct problems

We replicate the entire six-method ladder on three further classic structures sharing nothing with visa allocation but the SPV + greedy/permutation method-

**Table 5.** Mean Friedman rank (lower is better; 30 paired seeds) on four structurally distinct problems. A permutation-native method (pm) is best on every structure (**bold**); the real-coded NSGA-II falls below random restart only on the selection landscapes (visa, knapsack). “rk” = random-key.

| Method | Enc. | Visa | Knapsack | TSP | Flow-shop |
|---|---|---|---|---|---|
| perm-NSGA-II | pm | **1.60** | **1.10** | 2.00 | **1.80** |
| perm-MOEA/D | pm | 2.53 | 2.87 | **1.00** | 2.83 |
| Discrete-MOHHO | pm | 2.23 | 2.10 | 4.00 | 2.63 |
| MOHHO (real-coded) | rk | 4.67 | 3.93 | 5.00 | 5.07 |
| Random restart | rk | 4.20 | 5.23 | 6.00 | 5.93 |
| NSGA-II (real-coded) | rk | 5.77 | 5.77 | 3.00 | 2.73 |

ology: a 0/1 multidimensional knapsack (MOMKP, a *selection* landscape), a tri-objective traveling salesman (mo-TSP, *sequencing*), and a tri-objective permutation flow-shop (mo-PFSP, *scheduling*); same budget, 30 common seeds, paired Friedman + Nemenyi (Table 5). One result is robust across all four structures: a permutation-native method is *always* the best optimizer, and neither blind random restart nor any real-coded method ever wins—matching operators to the representation reliably produces the winner, with the family second-order among matched methods (all four omnibus tests reject equality, $\chi^2 = 112$–$150$, $p < 10^{-21}$). Discrete-MOHHO is top-two on three of the four and the most stable matched method. What does *not* generalize is the dramatic “GA below random sampling”: it holds on the selection landscapes (visa, knapsack, where the real-coded NSGA-II is worst) but reverses on the sequencing landscapes (TSP, flow-shop, where it is competitive and random restart is worst instead). The mechanism: operator order-preservation is problem-independent ($\tau \approx 0.99$), but its *consequence* is landscape-dependent—near-identity SBX freezes search on selection landscapes yet acts as useful local search on sequencing ones. A headroom scalar does not capture which (Spearman $\rho = -0.90$, $n = 5$, inconclusive). The transferable law is thus precise: representation-matched operators win on every structure; the GA-collapse is selection-landscape-specific.

## 7 Discussion

*The trade-off is structural.* Two countries hold roughly three-quarters of demand with 10–13-year waits; the 7 % ceiling forbids fully serving them, so reducing disparity ($f_2$) leaves unserved load ($f_1 \uparrow$) and equitable spreading idles saturated categories ($f_3 \uparrow$). The three extreme solutions (Table 3) are mutually non-dominated—genuine three-way conflict. We keep $f_1$ as a distinct objective (its range is more than an order of magnitude narrower than $f_2$'s because demand 1,800,000 far exceeds the 140,000 visas) because it is the minimum-$f_1$ policy that dominates FIFO, which a two-objective reduction would obscure.

*What governs performance, and policy.* The ladder shows performance is governed by the feasibility-preserving decoder and the representation–operator match, not the metaheuristic family: on this (selection) landscape a tuned real-coded GA even falls below blind sampling (its operators are near-identity on the decoded permutation, $\tau = 0.99$), while permutation-native operators lift three paradigms to within about 1 % of one another—and across four structures a matched operator is always the best optimizer (Section 6.5). The guidance is to invest in the decoder and representation before the search heuristic, and to prefer the most stable optimizer (Discrete-MOHHO) for single-run decisions. For policy, the front is a quantified menu: reducing disparity from 13.0 to 2.0 years costs only $\sim$2.4 % in $f_1$ relative to FIFO [8,11]. Each front solution maps to a concrete allocation in which India and China stay at their 7 % ceiling while visas FIFO leaves in the residual pool are redirected to underserved nationalities.

*Limitations.* The model covers a single fiscal year, uses 21 representative countries, and assumes every assigned visa is used. On the data: the country-level totals are anchored to published aggregates (about 1,800,000 pending cases; India $\approx$ 63 %, China $\approx$ 14 %) [3,19,20], but the per-(country, category) split and priority dates are a plausible synthetic disaggregation (only India, China, the Philippines, and Mexico have category-level demand grounded in published data). These affect external validity, not the internal comparison among the six methods, which share identical data, decoder, and budget. *Ethically*, $f_2$ encodes one contestable notion of fairness and treats nationalities as allocation buckets; we therefore present the front as auditable decision support, never an autonomous decision-maker, and the per-country figures illustrate method behavior rather than recommend real allocations.

## 8 Conclusions

We formulated the annual allocation of U.S. employment-based visas as a three-objective integer program with six hard constraints and solved it through a feasibility-preserving SPV + greedy decoder that guarantees feasibility by construction—no penalties, no repair. A Taguchi L9 design replaced empirical parameter choices with a confirmed operating point. Across 30 independent runs the recovered Pareto front Pareto-dominates the FIFO incumbent—chiefly by eliminating waste while matching its waiting load and maximal disparity—with 82 zero-waste solutions. Our central result comes from a controlled six-method ladder: the decoder alone (random restart) already beats *both* real-coded searches (NSGA-II and the MOHHO swarm), and—the lesson—the representation–operator match, not the metaheuristic family, governs performance. Matching operators to the representation lifts three distinct paradigms (a GA, a decomposition method, and a Harris Hawks optimizer) to within about 1 % of one another and above every random-key method; the Discrete-MOHHO we introduce improves on classic MOHHO by 4.7 % ($p = 9.3 \times 10^{-10}$) and is the most stable optimizer of all (CV 0.54 %), within 0.5 % of the best. All conclusions hold across five perturbed-demand instances, and a four-structure study (knapsack, TSP, flow-shop) sharpens what generalizes: a representation-matched operator is the best optimizer on *every* structure and the family is second-order among matched methods (the robust law), whereas the collapse of a tuned GA below blind sampling occurs only on *selection* landscapes (visa, knapsack)—on *sequencing* landscapes (TSP, flow-shop) the same near-identity operators act as useful local search and the GA is competitive. A three-axis robustness analysis—non-saturating decoders, reference-point sweeps, and an exact MILP front—shows the representation effect is not an artifact of the decoder's budget saturation but only amplified by it, modest on this $f_2$-dominated instance and carried, as a general law, by the four structures. Order-preservation ($\tau$=0.99) is problem-independent; whether it freezes or refines is landscape-dependent, with the precise determinant ($\rho = -0.90$ at $n = 5$, inconclusive) left open. Beyond the numbers, the method yields an auditable menu of feasible trade-offs as decision support, not a policy prescription. Future work includes a multi-period formulation, the full country roster, further problem domains beyond the four studied here, and additional optimizers such as SPEA2.

**Data and Code Availability.** The problem data, the optimizer, the Taguchi design, and the scripts that reproduce every figure and table are available at https://github.com/jrebull/MIAAD_Harrisv2.

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